

Logic puzzles

THE FRUSTRATED FARMER

A farmer is trying to use a small boat to row a fox, a chicken, and a bag of corn across a river. However, he can only take one thing at a time in the boat. If he leaves the fox with the chicken, the fox will eat the chicken. If he leaves the chicken with the corn, the chicken will eat the corn. How can the farmer get across the river without anything eating anything else?

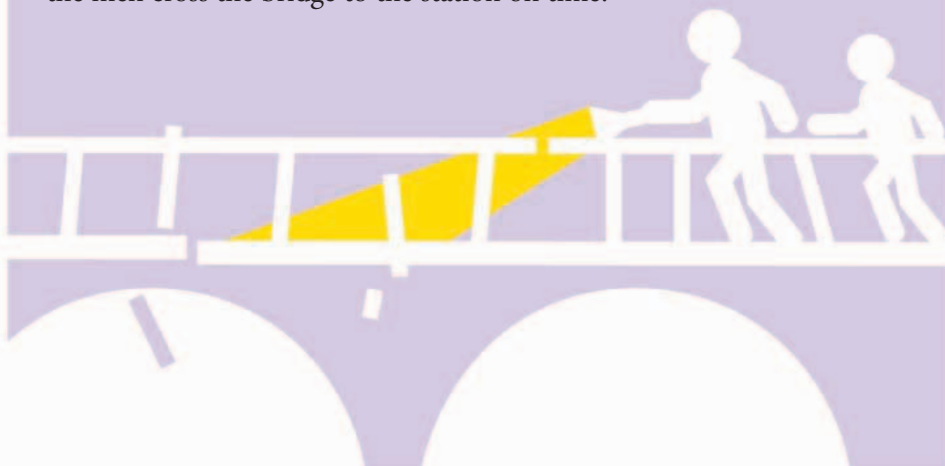


TWO AT A TIME

A group of four men—made up of two brothers plus their father and grandfather—is walking to a railway station in the dark. They come to an old, narrow bridge that leads to the station. The bridge can support just two people at a time and they have only one flashlight between them, so after one pair has crossed, one of the men will have to bring the flashlight back for the next pair. The four men take different times to cross the bridge.

- Brother one takes 1 minute
- Brother two takes 2 minutes
- The father takes 5 minutes
- The grandfather takes 10 minutes

Each pair can walk across the bridge only as fast as the slowest man, and the next train arrives at the station in 17 minutes. How can all the men cross the bridge to the station on time?



THE RIGHT DOOR?

A prisoner is given a chance to win his freedom. In his cell are two doors—behind one is a hungry lion and behind the other is the exit to the prison. In front of each door stands a guard—one guard always speaks the truth, and the other always lies. The prisoner is allowed to ask one of the guards just one question. So what question should he ask to gain his freedom?

THE PRISONER SHOULD ASK EITHER GUARD THIS QUESTION: "If I asked the other guard which is the door to freedom, which door would he point to?" The honest guard, knowing that the other guard would lie, would answer by pointing to the door with a lion behind it. The dishonest guard, knowing that the other guard would point to the good door, would answer by pointing to the door with the lion behind it. Either way the answer is the same: They would both point to the door with a lion behind it, and the prisoner should take the other door.

TWO AT A TIME
Brother one and brother two cross the bridge together, taking 2 minutes. Brother one returns, taking 1 minute. The father and grandfather cross together, taking 10 minutes. Brother two returns, taking 2 minutes, then brother one and brother two cross together, taking 2 minutes. $2 + 1 + 10 + 2 + 2 = 17$, so they should all get to the station just in time for the train.

THE FRUSTRATED FARMER
The farmer crosses the river first with the chicken, and leaves it on the other side. He then returns, picks up the fox, and crosses again. Then he swaps the fox for the chicken, so that they are not left together, and takes the chicken back. He then swaps the chicken for the grain, and takes the grain across, leaving it with the fox. He then returns, picks up the chicken, and takes it to the other side.

Logic puzzles

Number puzzles
1 Three, 2 e, 3 a, 4 (each pair of numbers adds up to the next one in the sequence), 5 b, 6 One—the first one! 7 This is because 1 hour 20 minutes is the same as 80 minutes, 8 e, 9 d, 10 d, 11 c, 12 d, 13 Just the four dead crows. The rest flew away when they heard the gunshots, 14 b, 15 b (all the numbers are square numbers), 16 None—it was Noah who built the Ark, not Moses! 17 Two—they're the ones you took! 18 Just over an hour.

ANSWERS

